Metrics & Alerting

- Data & Output Quality
- System and Runtime
- Agent Behavior
- Governance and Risk
- Cost

session
turn
orchestrator
subagent
tool
handoff
hitl
policy
base

Data & Output Quality

Input Data Quality

Logged Field	Metrics	Alert Signal
turn user_message	• Input length distribution • Null/empty rate	Drift vs 30 day baseline
turn retrieved_context	• Retrieval hit rate • Context volume per turn • Empty retrieval rate	Spike in empty retrievals
turn tokens_prompt	• Prompt token distribution • Input token drift	drift > threshold vs baseline

	• Mean/StdDev over time	

Model Performance

Logged Field	Metrics	Alert Signal
turn groundedness_score	• Groundedness mean/StdDev • % below threshold • Groundedness by use_case	Mean drops below threshold
turn outcome	• Acceptance rate • Rephrasing rate • Correction rate	Acceptance rate declines >X%
turn *subagent* subagent_version outcome	• Version-over-version quality delta • Regression detection	Quality drop

Agent Output Quality

Logged Field	Metrics	Alert Signal
turn agent_response outcome	• Response null rate • Acceptance rate • Rephrasing rate	Null Response Spike
turn user_correction	• Correction rate	Correction rate up trend
subagent degraded_response	• Degraded response rate • Degradation by subagent_id	Degraded rate > threshold

	• Degradation by use_case	
subagent output output_size_bytes	• Output size distribution • Output size drift	Output size drift

LLM Signals

Logged Field	Metrics	Alert Signal
turn *subagent* tokens_completion	• Response length distribution • Output token drift • Entropy changes over time	Sudden concentration or collapse
turn *subagent* model_used outcome	• Model distribution • Output quality by model • Model mix drift	Unexpected model shift
turn safety_triggered	• Refusal rate • Safety trigger rate by use_case	Refusal rate spike

Retrieval Health

Logged Field	Metrics	Alert Signal
turn retrieved_context	• Retrieval hit rate • Cache hit rate on retrieval	
tool cache_hit	• Cache miss trend	Cache hit rate drops

tool output_payload	• Tool output quality • Not-found rate by tool_id	Not-found rate increases

System and Runtime

Real-time Serving

Logged Field	Metrics	Alert Signal
turn latency_ms_total	• End-to-end latency • Latency by use_case / channel • SLA breach rate	latency > threshold
session final_outcome error	• Session error rate • Error rate by channel	Error rate exceeds threshold
tool success failure_reason	• Tool error rate • Error type breakdown (timeout, auth, not_found) • Error rate by tool_id	Tool error rate spikes

Agent Runtime

Logged Field	Metrics	Alert Signal
orchestrator latency_ms	• Orchestrator decision latency • Latency contribution to turn	Latency exceeds threshold

	total • Slow-decision alert	
subagent latency_ms	• Latency by subagent_id • Subagent share of turn latency	Latency exceeds threshold
tool success failure_reason	• Tool call latency • Slow tool by tool_id • Tool latency share of turn	Tool error rate spikes Latency exceeds threshold
orchestrator *subagent* *tool* retry_count	• Retry rate by layer • Max-retry breach rate	Retry rate increases
orchestrator fallback_triggered	• Fallback rate • Fallback by subagent • Fallback cost impact	Fallback rate spikes

Batch Execution Health

subagent success + failure_reason	• Success / failure rate • Step-level failure visibility • Retry rate	Failure rate exceeds threshold
subagent end_utc - subagent_start	• Runtime duration • Duration trend over time	Runtime duration anomaly
tool end_utc - tool_start	• Tool execution duration • Slow tool detection	

Integration Reliability

Logged Field	Metrics	Alert Signal
handoff acknowledgement_received	• Delivery success rate • Unacknowledged handoff rate • Delivery SLA compliance	Unacknowledged handoff alert
handoff destination_type destination_id	• Handoff volume by destination • Downstream team load	
tool success error_type	• External API success rate • Auth error rate • Timeout rate by integration	Auth errors or timeout spike

Agent Behavior

Agent Behavior

Field	Metrics	Alert
goal_completion_accuracy	% of outputs meeting intent	<90%
hallucination_rate	% of incorrect/unsupported responses	>2%
constraint_violation_rate	% violating business rules/poicies	>0%

relevance_score	semantic quality of response	<{baseline}

Routing and Decision Quality

Logged Field	Metrics	Alert Signal
orchestrator decision_type	• Route distribution (route vs respond vs escalate) • Fallback rate • Direct-response rate	Escalation rate spikes
orchestrator confidence_score	• Mean routing confidence • Low-confidence rate • Confidence vs outcome correlation	Low-confidence rate increases
orchestrator candidate_options	• Route diversity • Underutilized subagent detection • Routing concentration	Single-subagent concentration
orchestrator decision_index	• Avg decisions per turn • Multi-hop frequency • Routing complexity trend	Complexity trend increasing
orchestrator orchestrator_version	• Version-over-version routing quality	Quality drop post-deploy

	• Regression detection post-deploy	

HITL Operations

Logged Field	Metrics	Alert Signal
hitl review_start/end_utc review_duration_ms	• Queue wait time • Review duration • Reviewer throughput	Queue SLA breach
hitl sla_breached	• SLA breach rate by escalation_type • Breach trend over time	SLA breach rate alert
hitl reviewer_id	• Inter-rater reliability • Per-reviewer override rate • Reviewer consistency	Outlier reviewer detected

Session Outcomes

Logged Field	Metrics	Alert Signal
session final_outcome	• Completion rate • Abandonment rate • Escalation rate • Error rate	Queue SLA breach

session total_turns	• Avg turns per session • Turn distribution • High-turn session rate	SLA breach rate alert
session hitl_count	• Automation rate (hitl=0 %) • HITL volume trend	Override rate spikes
turn resumed_from_hitl hitl_event_id	• Post-HITL completion rate • HITL-to-completion funnel	Outlier reviewer detected

Escalation Patterns

Logged Field	Metrics	Alert Signal
hitl escalation_type	• Escalation breakdown by type • Policy vs confidence vs safety split • Escalation trend by use_case	Safety escalations spike
hitl escalation_reason	• Escalation cause taxonomy • Frequent escalation patterns • Top trigger text clusters	
hitl escalating_agent_id	• Escalation rate by subagent • High-escalation agent alert	Subagent escalation rate high

Governance and Risk

Policy and Safety

Logged Field	Metrics	Alert Signal
policy policy_type severity action_taken	• Policy fire rate by type • Severity distribution • Block vs mask vs flag rate • Critical event rate	Critical severity event fires
policy trigger_reason	• Rule/classifier hit frequency • Top trigger taxonomy • False positive candidates	
turn safety_triggered	• Safety trigger rate • Safety events per session • Safety rate by channel	Safety rate spikes
policy policy_type (prompt_injection)	• Prompt injection attempt rate • Injection rate by channel	Injection rate above threshold
policy policy_type (pii_detected)	• PII detection rate • PII by channel / use_case	PII spike alert

HITL Standards Compliance

Logged Field	Metrics	Alert Signal
hitl escalation_type	• Escalation coverage by type • % sessions with HITL when required	Required HITL not triggered
hitl sla_breached reviewer_decision	• SLA compliance rate by escalation_type • Reviewer decision audit trail	SLA compliance drops
hitl override_instruction	• % overrides with documented reason	Undocumented overrides

Model Lifecycle and Versioning

orchestrator orchestrator_version	• Active version inventory • Version distribution across traffic • Deprecated version usage	Deprecated version in prod
subagent subagent_version	• Subagent version inventory • Version drift across sessions	
subagent model_used	• Active model inventory • Unapproved model detection	Unapproved model detected

tool tool_version	• Tool version inventory • Old tool version usage	Deprecated tool version

Data Lineage and Retention

Logged Field	Metrics	Alert Signal
subagent input_context (redacted)	• PII redaction compliance rate • Redaction failure detection	Redaction failure
turn retrieved_context	• Source traceability rate • Context source logging completeness	
session session_start/end_utc user_id	• Data retention window compliance • Session log completeness	Retention policy breach

Cost

Attribution

Field	Metrics	Alert
subagent tokens_prompt tokens_completion model_used	• LLM token cost by model • Cost by subagent_id	

	• Batch vs real-time cost split	
base environment	• Dev / staging / prod cost split • Experimental vs production spend	Dev spend exceeds cap
session base use_case agent_id	• Cost per use case • Cost per agent / team	
tool tokens_used	• LLM-backed tool cost • Tool AI spend vs direct spend	

Unit Economics

Logged Field	Metrics	Alert Signal
turn tokens_prompt tokens_completion	• Cost per turn • Cost per session • Cost per user interaction	Cost per interaction trend up
turn *subagent* subagent_count tool_calls_made	• Cost per routing hop • Cost per tool call • Multi-hop cost concentration	
subagent model_used tokens_*	• Cost per token by model • Model cost efficiency ratio	

Waste and Inefficiency

Logged Field	Metrics	Alert Signal
orchestrator *subagent* *tool* review_start/end_utc review_duration_ms	• Retry cost overhead • Wasted spend from retries • Retry rate trend	Retry cost overhead spikes
orchestrator sla_breached	• Fallback cost impact • Fallback vs primary cost delta	
tool reviewer_decision	• Cost avoidance from cache • Cache miss cost • Cache ROI	Cache hit rate drops
subagent reviewer_id	• Cost of degraded responses • Waste from failed subagent runs	

Guardrails and Forecasting

Logged Field	Metrics	Alert Signal
turn tokens_prompt tokens_completion	• Token growth modeling • Projected cost at traffic growth • Spend anomaly detection	Sudden token usage spike
turn subagent_count tool_calls_made	• Complexity growth trend • Routing cost forecast	

base *session* environment use_case	• Budget utilisation by env • Spend cap monitoring	Budget threshold breached